MMDU MULLANA OFFICIAL SYLLABUS • BPHY-001 • UNIT 4

Unit 4: Crystal Structures & Free Electron Theory of Metals (Solid State Physics)

Subject: Applied Physics (BPHY-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 40 mins

Space lattice, basis, 7 crystal systems & 14 Bravais lattices, SC/BCC/FCC derivations (coordination number, atomic radius, and atomic packing factor APF), 12-point Schottky vs Frenkel defects comparison, Classical Drude theory & specific heat failures, Sommerfeld quantum theory, Fermi-Dirac distribution, Fermi energy derivation at 0 K, and Richardson-Dushman thermionic emission equation.

1. Crystallographic Fundamentals & 14 Bravais Lattices

1. Lattice Points, Basis & Crystal Structure

- **Space Lattice:** An infinite periodic 3D geometric array of mathematical points where every point has identical spatial surroundings.

- **Basis (Motif):** A group of one or more atoms structurally associated with each lattice point.

> $$\text{Crystal Structure} = \text{Space Lattice} + \text{Basis}$$

2. The 7 Crystal Systems

Crystal System	Axial Lengths	Interaxial Angles	Bravais Lattices Count
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	3 (P, I, F)
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	2 (P, I)
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	4 (P, C, I, F)
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	2 (P, C)
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	1 (P)
Trigonal (Rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	1 (P)
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	1 (P)

| **Total** | | | **14 Bravais Lattices** |

2. Cubic Crystal Geometries: SC, BCC & FCC APF Derivations

Complete Step-by-Step Derivation of Cubic Geometries

$$\text{Atomic Packing Factor (APF)} = \frac{n_{\text{eff}} \times \frac{4}{3}\pi r^3}{a^3}$$

Parameter	Simple Cubic (SC)	Body-Centered Cubic (BCC)	Face-Centered Cubic (FCC)
Effective Atoms (n_{eff})	$8 \times \frac{1}{8} = \mathbf{1}$	$8 \times \frac{1}{8} + 1 = \mathbf{2}$	$8 \times \frac{1}{8} + 6 \times \frac{1}{2} = \mathbf{4}$
Coordination Number (CN)	**6**	**8**	**12**
Touching Geometry	Edge: $2r = a$	Body diagonal: $4r = \sqrt{3}a$	Face diagonal: $4r = \sqrt{2}a$
Atomic Radius (r)	$r = \frac{a}{2}$	$r = \frac{\sqrt{3}}{4}a$	$r = \frac{\sqrt{2}}{4}a = \frac{a}{2\sqrt{2}}$
Nearest Neighbor (d)	$d = a$	$d = \frac{\sqrt{3}}{2}a$	$d = \frac{a}{\sqrt{2}}$
APF Proof	$\frac{1 \times \frac{4}{3}\pi(a/2)^3}{a^3} = \frac{\pi}{6}$	$\frac{2 \times \frac{4}{3}\pi(\frac{\sqrt{3}a}{4})^3}{a^3} = \frac{\sqrt{3}\pi}{8}$	$\frac{4 \times \frac{4}{3}\pi(\frac{a}{2\sqrt{2}})^3}{a^3} = \frac{\sqrt{2}\pi}{6}$
Packing Efficiency	**52.4%**	**68.0%**	**74.0% (Close-Packed)**

| **Classic Elements** | Polonium (α-Po) | Fe, Cr, W, Mo, Na | Cu, Al, Au, Ag, Pt |

3. Crystal Defects: Schottky vs. Frenkel Defects (12-Point Analysis)

12-Point Comparative Matrix: Schottky vs. Frenkel Defects

#	Dimension	Schottky Defect	Frenkel Defect
1	**Fundamental Mechanism**	**Pair Vacancy:** Equal number of cations and anions missing from normal lattice sites.	**Dislocation:** An ion (usually smaller cation) is displaced to an interstitial site.
2	**Stoichiometry**	Crystal remains strictly stoichiometric and electrically neutral.	Crystal remains strictly stoichiometric and electrically neutral.
3	**Ion Radius Ratio**	Occurs when cation and anion are of **comparable size** ($r_+/r_- \approx 1$).	Occurs when there is a **large size difference** ($r_+ \ll r_-$).
4	**Coordination Number**	Favored in crystals with **high coordination number** ($CN = 6$ or 8).	Favored in crystals with **low coordination number** ($CN = 4$).
5	**Effect on Density**	Lattice mass decreases for constant volume $\implies$ **Density decreases**.	No ions exit the crystal lattice $\implies$ **Density remains unchanged**.
6	**Dielectric Constant**	Negligible effect on dielectric permittivity.	**Dielectric constant increases** due to charge crowding at interstitials.
7	**Lattice Strain**	Minimal localized strain relaxations.	Causes high internal localized lattice strain.
8	**Equilibrium Concentration**	$n_s \approx N \exp\left(-\frac{E_s}{2k_BT}\right)$	$n_f \approx \sqrt{NN_i} \exp\left(-\frac{E_f}{2k_BT}\right)$
9	**Transport Mechanism**	Vacancy hopping conductivity.	Interstitial ionic diffusion conductivity.
10	**Entropy Contribution**	Increases configurational pair entropy.	Increases vacancy-interstitial entropy.
11	**Representative Crystals**	**NaCl, KCl, KBr, CsCl**	**AgCl, AgI, ZnS**

| **12** | **Unique Special Case** | **Silver Bromide (AgBr)** exhibits **BOTH** Schottky and Frenkel defects depending on temperature! |

4. Free Electron Theory of Metals: Sommerfeld Quantum Theory & Thermionic Emission

1. Classical Theory (Drude-Lorentz) & Its Severe Failures

- **Electrical Conductivity:** $\sigma = \frac{ne^2\tau}{m}$
- **Wiedemann-Franz Law:** $\frac{K}{\sigma T} = \frac{3}{2}\left(\frac{k_B}{e}\right)^2 = L$
- **Classical Failures:**

1. *Specific Heat Catastrophe:* Predicts electronic specific heat $C_{v\,\text{el}} = \frac{3}{2}R$. Observed value is $\sim 0.01R$ at room temperature (overestimated by a factor of 100!).

2. *Paramagnetism:* Predicts Curie temperature dependence $\chi \propto 1/T$, but metal electron paramagnetism is experimentally independent of temperature.

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2. Quantum Free Electron Theory (Sommerfeld) & Fermi-Dirac Distribution

Sommerfeld treated electrons as quantum particles obeying the **Pauli Exclusion Principle** and **Fermi-Dirac Statistics**:

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$$

- **At $T = 0$ K:**

- For $E < E_F$: $f(E) = 1$ (all quantum states filled).
- For $E > E_F$: $f(E) = 0$ (all quantum states vacant).

- **At $T > 0$ K:** Thermal energy smears occupancy within $\sim k_B T$ of E_F. At $E = E_F$:

$$f(E_F) = \frac{1}{1+1} = 0.5 \quad (50\% \text{ probability})$$

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3. Derivation of Fermi Energy E_F at 0 K

The 3D density of quantum states with electron spin degeneracy 2 is:

$$g(E)dE = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2} dE$$

Integrating up to $E_F(0)$ for total electron count N:

$$N = \int_0^{E_F(0)} g(E)dE = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} [E_F(0)]^{3/2}$$

Solving for $E_F(0)$ in terms of electron density $n = N/V$:

$$E_F(0) = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$$

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4. Thermionic Emission: Richardson-Dushman Equation

Thermal ejection of electrons when thermal kinetic energy exceeds the work function barrier ϕ:

> $$J = AT^2 e^{-\phi/(k_B T)}$$

- Universal Richardson constant: $A = \frac{4\pi m e k_B^2}{h^3} \approx 1.2 \times 10^6 \ \mathrm{A \cdot m^{-2} \cdot K^{-2}} = 120 \ \mathrm{A/cm^2 K^2}$.
- ϕ: Cathode work function (2 to 5 eV).